

PathFinder: AI-Powered Career & Educational Advisor

Abstract

In today's rapidly evolving job market, students and professionals face significant challenges in identifying suitable career paths and finding the right educational resources to bridge their skill gaps. Traditional career guidance systems are often static, generic, and fail to provide personalized, actionable roadmaps. To address this, the proposed project, **PathFinder**, is designed as an intelligent, AI-powered Career and Educational Advisor system. PathFinder integrates advanced Natural Language Processing (NLP), Machine Learning, and Large Language Models (LLMs) to deliver highly personalized career recommendations and learning roadmaps. The system allows users to enter their profile details manually or upload their resumes in PDF or Word format. The background engine parses the resume, extracts key skills using a hybrid approach combining SpaCy, regular expressions, and the Groq API (powered by Llama-3), and maps them to standard skill taxonomies. PathFinder then utilizes TF-IDF vectorization and cosine similarity, alongside dynamic LLM prompting, to recommend the most compatible career options and calculate a match percentage. Furthermore, it leverages a ChromaDB vector database containing indexed courses and certifications to retrieve semantically relevant educational resources. Finally, the system generates a tailored, week-by-week learning roadmap and calculates a 2030 Future-Proofing Index to guide users through their upskilling journey. With a simple and user-friendly interface, PathFinder supports continuous skill tracking and maintains analysis history for future reference.

Keywords

Artificial Intelligence, Career Advisory, Resume Parsing, Natural Language Processing, TF-IDF Vectorization, Semantic Search, ChromaDB, Future-Proofing, Upskilling Roadmaps

Problem Statement

Students and job seekers often encounter challenges in identifying suitable career paths that align with their existing skills, detecting critical skill gaps, and selecting relevant educational resources efficiently. Traditional manual career counseling can be expensive, time-consuming, and may not provide objective or data-driven insights. Moreover, the overwhelming abundance of online courses makes it difficult for learners to chart a structured learning path. Therefore, an intelligent automated system is required to analyze candidate profiles effectively and provide practical recommendations for career alignment and improved resource utilization.

Objectives

- To extract candidate skills and qualifications automatically from uploaded resumes (PDF/DOCX) using AI and NLP techniques.
- To assess crop health and career compatibility using hybrid matching algorithms (TF-IDF, Cosine Similarity, and Llama-3 LLM).
- To detect skill gaps between candidate profiles and target career requirements.
- To retrieve semantically relevant courses and professional certifications using vector search via ChromaDB.
- To generate structured, week-by-week personalized learning roadmaps for skill acquisition.
- To analyze skill resiliency against AI automation using a 2030 Future-Proofing Index.
- To maintain profile analysis records for future monitoring and career decision-making.

Proposed System

The proposed system is an intelligent career and educational guidance application that integrates Artificial Intelligence, Natural Language Processing, and Vector Databases to analyze candidate profiles. Users can input their details manually or upload a resume file through the application, after which the system processes the document to extract skills, evaluate career compatibility, and identify missing skill gaps. Based on the analysis, the system generates appropriate recommendations for courses, professional certifications, week-by-week learning roadmaps, and 2030 future-proofing resiliency. The application is designed to be simple, accessible, and useful for students, job seekers, and professionals.

Methodology

1. The user inputs profile details manually or uploads a resume document (PDF/DOCX) through the application.
2. The system preprocesses the document to extract raw text and enhance data quality.
3. Artificial Intelligence and Natural Language Processing models analyze the text to detect skills, educational background, and interests.
4. The condition and career compatibility of the candidate is evaluated based on TF-IDF and Cosine Similarity analysis results.
5. The system generates suitable recommendations for courses, certifications, upskilling roadmaps, and future-proofing.
6. The analysis results are stored in a local database for future reference and continuous career monitoring.

Key Features

- Resume file upload (PDF/DOCX) and automated NLP skill analysis.
- Identification of career paths using AI-based matching techniques.
- Evaluation of overall career compatibility and match percentage.
- Detection of target skill gaps with visual indicator checklists.
- Smart recommendations for courses, certifications, and educational resources.
- Personalized week-by-week upskilling roadmap generation.
- 2030 Future-Proofing Index evaluating skill resilience against AI automation.
- Direct Job Board integration with skill alignment badges.
- Simple and user-friendly web interface.
- Storage of analysis history for future reference and decision-making.

Expected Outcome

The expected outcome of this project is the development of an intelligent and easy-to-use application that assists users in monitoring their career readiness and managing upskilling resources more effectively. The system is expected to support early detection of skill gaps, reduce wastage of learning time, improve candidate productivity, and promote effective, structured professional growth.

Conclusion

The project **PathFinder: AI-Powered Career & Educational Advisor** presents a technology-based approach for improving career guidance and resource management in education. By applying Artificial Intelligence, Natural Language Processing, and Vector Search, the system can help users identify skill gap issues and make better career decisions. Overall, the proposed solution contributes to sustainable professional development by enhancing learning productivity and reducing unnecessary resource consumption.